Solution:

Let the intercepts cut off from the axes be of lengths a and $-a$.

$\therefore$ Equation of the line is of the form

$$\frac{x}{a} - \frac{y}{a} = 1 \Rightarrow\ x - y = a.$$

Since it passes through $(-1, 1)$

$\Rightarrow (-1) - (1) = a \Rightarrow a = -2.$

Equation of the line is $x - y + 2 = 0$.

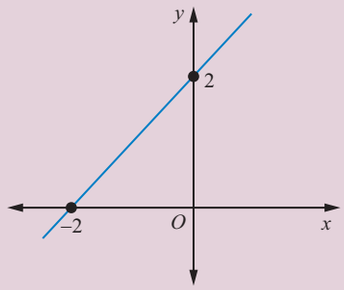


Figure 6.30

Example 6.15 A straight line L with negative slope passes through the point $(9,\ 4)$ cuts the positive coordinate axes at the points P and Q. As L varies, fin the minimum value of $|OP| + |OQ|$, where O is the origin.

Solution:

Let m be the slope of the line L. Since it passes through the point $(9,\ 4)$ the equation of the line L is $y - 4 = m(x - 9)$.

The points P and Q are respectively $(9 - \frac{4}{m},\ 0)$ and $(0,\ 4 - 9m)$.

$$\begin{aligned}
(m < 0)\ |OP| + |OQ| &= \left|9 - \frac{4}{m}\right| + |4 - 9m| \\
&= \left|9 + \frac{4}{k}\right| + |4 + 9k| \quad (m < 0,\ \text{take } m = -k,\ k > 0) \\
&= (9 + \frac{4}{k}) + (4 + 9k) \quad \text{(all terms are positive)} \\
&= (4 + 9) + (\frac{4}{k} + 9k\ ) \\
&\geq 13 + 2\sqrt{\frac{4}{k} \times 9k} \quad (\textit{A}\text{rithmetic mean} \geq \textit{G}\text{eometric mean}) \\
|OP| + |OQ| &\geq 25
\end{aligned}$$

Therefore, the minimum absolute value of $|OP| + |OQ|$ is 25.

Example 6.16 The length of the perpendicular drawn from the origin to a line is 12 and makes an angle $150°$ with positive direction of the x-axis. Find the equation of the line.

Solution:

Here, $p = 12$ and $\alpha = 150°$, So the equation of the required line is of the form

$$\begin{aligned}
x\cos\alpha + y\sin\alpha &= p \\
\text{That is, } x\cos 150° + y\sin 150° &= 12 \\
\Rightarrow \sqrt{3}x - y + 24 &= 0
\end{aligned}$$